

Arnab Ghosh

Portfolio

GitHub

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EDUCATION

Vellore Institute of Technology Vellore

Bachelor of Technology – Information Technology: CGPA: 8.48/10

Vellore, India

2021 – 2025

EXPERIENCE

Software Engineer

Firstsource Solutions Limited

Jul 2025 – Present

Remote

- Built and deployed a **GenAI-powered intelligent document automation system** for Deed, Mortgage, and Tax processing — **integrating LLM solutions into production software workflows** via end-to-end pipelines for ingestion, prompt-engineered extraction, structured JSON transformation, and enterprise integration.
- Integrated REST API-based workflows with Resware, eliminating manual processing and reducing package handling time to ~4.5 minutes with **100% extraction accuracy**.
- Implemented **Guardrails-based schema enforcement** with Pydantic validation, automated retry/correction mechanisms, and **custom HuggingFace transformer validation layers** for semantic verification, hallucination detection, and confidence scoring — reducing structured extraction failures by over **40%**.
- Developed an **enterprise AI guardrails framework** with 10+ content validators (jailbreak, prompt injection, PII, toxicity, bias) across **Azure, AWS, and GCP** safety services with primary/fallback model design, and authored cross-cloud cost estimation guiding production deployment.
- Engineered a **RAG pipeline over 100K+ documents** for automated form testing and validation, implementing chunking, vector indexing, and retrieval workflows to support scalable document search.
- Architected a **distributed batch inference system on Azure Batch** for large-scale claim processing — an event-driven **Azure Function coordinator** (Service Bus triggered) submits tasks to an autoscaling compute pool, with deterministic idempotent task IDs guaranteeing exactly-once processing under at-least-once delivery.
- Designed **multi-LLM load balancing across multiple Azure OpenAI endpoints** with per-task endpoint assignment and token-bucket **TPM throttling**, eliminating 429 rate-limit failures and sustaining high-throughput concurrent inference.
- Developed **Python automation tools using Playwright and n8n** for end-to-end form workflows — Excel data ingestion, automated form filling across authenticated sessions, and structured output generation.

PROJECTS

Travel Buddy — Python, FastAPI, Gemini 2.5 Flash, SQLite, yt-dlp, Docker

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AI-First Travel Companion with Multi-Modal Extraction Pipeline

- Designed and developed an end-to-end application featuring a **4-stage multi-modal extraction pipeline** (yt-dlp metadata → Gemini video → Gemini audio → Open Graph fallback) with graceful degradation at every tier, converting Instagram reels, YouTube shorts, and Maps links into structured travel notes.
- Integrated **Gemini 2.5 Flash via REST API** using native vision, audio, and video modalities in a single call, with **responseSchema**-enforced JSON output requiring zero post-processing; debugged and hardened subprocess orchestration with timeouts, fallbacks, and stderr diagnostics.
- Designed the backend as **RESTful services across 9 route modules** over FastAPI/SQLite with idempotent, backward-compatible schema migrations; built a real-time GPS auto-visit detector and smart-routing endpoint (Haversine + nearest-neighbor + 2-opt heuristic); containerized with Docker and deployed to cloud.

AI Resume Editor — Python, FastAPI, OpenAI Function Calling, LaTeX, tectonic

GitHub

Agentic Resume-Tailoring System with LaTeX-Native Edit Pipeline

- Designed an **OpenAI function-calling agent loop** (up to 8 tool-call steps) with 7 tools spanning section retrieval, LaTeX validation, and staged-edit proposals for both single-section and multi-section editing modes.
- Enforced a server-side re-validation gate so every agent-proposed edit is independently **tested for LaTeX correctness and structural integrity** before reaching the user — including a custom guard against the model silently dropping section headings.
- Built a sandboxed **LaTeX compilation pipeline** (tectonic subprocess, timeout-bound, network-restricted) with a fast pre-compile syntax validator catching most errors in milliseconds instead of a multi-second recompile.

SKILLS

Programming Languages: Python, Java, JavaScript, SQL

Application Development: FastAPI, NodeJS, ExpressJS, REST APIs, ReactJS, Application Design, Debugging & Troubleshooting, Exception Handling

Databases: MongoDB, MySQL, SQLite, AWS RDS

GenAI & Automation: LLM Integration, Agentic AI, RAG, Prompt Engineering, HuggingFace Transformers, Guardrails, Playwright, n8n

Tools & Platforms: Docker, Azure (Batch, Functions, OpenAI, Service Bus), AWS, Git, Linux, Postman